Practice Problem Set

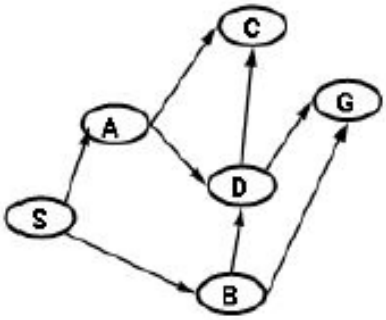
Solving Problems by Searching

1. For the simple vacuum-cleaner world with two locations, each of which may be clean or dirty, and with the agent being in one of the two locations, how many distinct environment states are possible? Create a state space diagram for this given scenario.

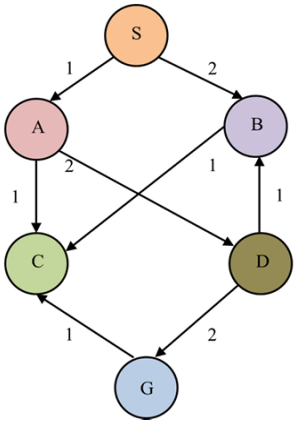
2. An agent controls the elevator in a 10-story building. On each floor, the doors can be open or closed. The elevator can also be moving between floors. How many states could the agent be in? Also identify PEAS description and environment type.

3. For each of the following graph search strategies, work out the order in which states are expanded, as well as the path returned by graph search. In all cases, assume ties resolve in such a way that states with earlier alphabetical order are expanded first. The start and goal state are S and G respectively, unless specified otherwise.
   - Breadth-First Search (with early goal testing and also with late goal testing; compare results)
   - Depth-First Search
   - Iterative Deepening Search
   - Uniform Cost Search

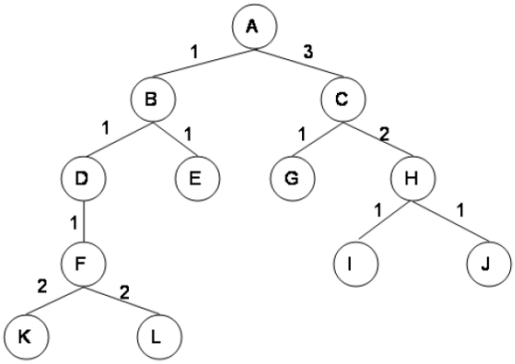
   a.



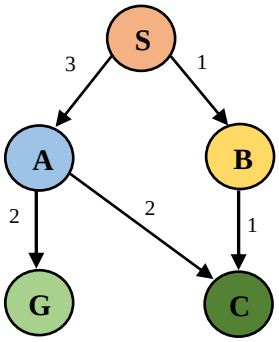
   b.



   c. Start state: A

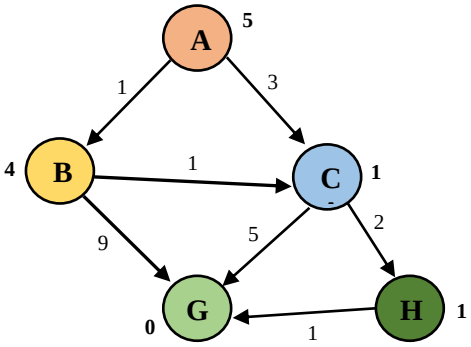


   d.

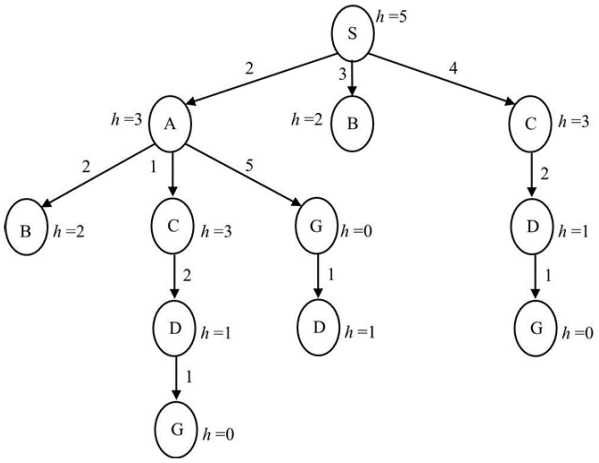


4. For each of the following graph search strategies, work out the order in which states are expanded, as well as the path returned by graph search. In all cases, assume ties resolve on first-come-first-served basis. The start and goal state are S and G respectively, unless specified otherwise.
   - Greedy search with the heuristic h shown on the graph.
   - A⋆ search with the same heuristic

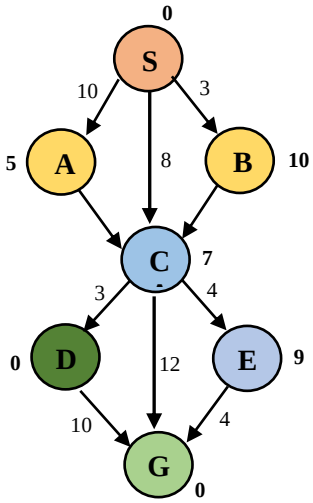
   a. Start state: A



b.

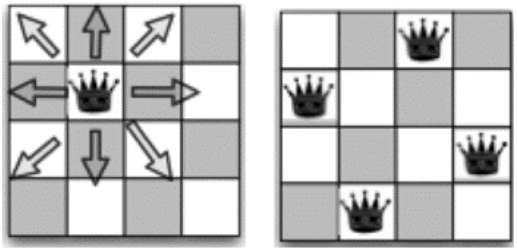


c.



5. For the graph given in problem 4.a, prove that all heuristics are admissible and consistent.

6. The n-Queens problem asks you to place 8 queens on a standard chessboard so that no two queens attack each other (i.e., no two queens share the same row, column, or diagonal).
   The figure below illustrates the simpler 4-Queens problem. On the left, arrows show that queens can move any number of squares along rows, columns, and diagonals. On the right is an example solution where no queens attack each other.



Formulate the 8-Queens problem as a search problem using the following guidelines:

1. **State Representation**: How will you represent a state of the board? Assume at most one queen per column.
2. **Start State**: Define what the initial board looks like. Assume any valid configuration with 8 queens placed.
3. **Successor Function**: Explain how to generate new states from a given state. Assume a queen can only be moved in its own column. How many successors are possible for this move?
4. **Goal Test**: Define how to check if a state is a solution.
5. **Heuristic Function (h)**: Define a heuristic that estimates how close a state is to the goal. (Hint: use the number of pairs of queens attacking each other.)

For your assumed start state:

- Draw the initial board, and compute its heuristic value h.
- Select two random successor states (by moving a queen in its column). For each, draw the board and calculate the new heuristic value.

7. The 15-Puzzle (sliding puzzle) consists of a 4×4 board containing 15 numbered tiles and one empty square. A legal move slides a tile adjacent to the blank into the blank. The goal is to reach the ordered configuration as shown below.



Formulate the 15-Puzzle as a search problem. Clearly state:

1. **State representation:** How will you represent a board state?
2. **Start state:** What is an initial configuration. Assume any solvable configuration)?
3. **Successor function:** How are successor states generated?
4. **Goal test**: How do you check whether a state is a goal?

Define the following heuristics and compute their values on you assumed initial state:

- **Misplaced-tiles heuristic:** Number of tiles not in their goal position.
- **Manhattan-distance heuristic**: Sum, over all tiles, of the Manhattan distance from the tile's current position to its goal position (blank ignored).

For each heuristic above, explain why it is admissible.
Also, comment on consistency (monotonicity) of each heuristic and which heuristic is generally more informed for A* on the 15-Puzzle. Explain why.